

# Ethan Radecki

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## Professional Summary

Data scientist and analyst with dual undergraduate degrees in Sport Analytics and Economics and a Master's in Applied Data Science (AI concentration). Experienced in statistical modeling, machine learning, and NLP, with hands-on work developing predictive models and testing hypotheses using Python, R, and SQL. Demonstrated ability to conduct research on real-world datasets and communicate technical findings clearly, committed to using data to drive decisions in sports, finance, and business strategy, while contributing to collaborative, problem-solving teams.

## Education

### **Syracuse University, Syracuse, NY**

David B. Falk College of Sport and Human Dynamics

Maxwell School of Citizenship and Public Affairs

Syracuse University School of Information Studies

B.S. in Sport Analytics, B.A. in Economics, M.S. in Applied Data Science

Undergraduate Graduation – May 2025

Graduate Graduation – May 2026

Cumulative GPA: 3.83 ~ *Magna Cum Laude*

## Experience

### **Syracuse University Stevenson Education Center, Front Desk Assistant**

September 2024 – May 2025

- Managed front desk operations requiring strong organizational skills, problem-solving, and effective communication with student populations.

### **Syracuse University Men's Lacrosse, Data Analytics & Sports Science Intern**

September 2023 – May 2025

- Designed and implemented quantitative analyses of biometric GPS performance data (speed, direction changes, jump height, heart rate) using statistical methods to identify patterns and inform player performance optimization.
- Built Power BI and Tableau dashboards, translating technical biometric metrics into actionable insights for coaches and the Head of Strength & Conditioning.
- Partnered directly with coaching staff as an internal stakeholder to deliver data-driven recommendations for practice planning and player-specific training routines.

### **Syracuse University Sport Analytics – Data in Italian Football, Student Intern**

January 2023 – April 2023

- Collaborated with student group to study Serie A analytics evolution through weekly discussions, readings, and analytical projects, synthesizing academic literature with industry practices through systematic analysis.
- Traveled to Italy for 10 days visiting four Serie A clubs, engaging with sports science and analytics teams to understand real-world applications of data science in professional football, culminating in a presentation of research findings and industry insights to senior data analyst leaders at AC Milan.

## Relevant Classes and Projects

### **Financial Analytics, Quant Reasoning, Deep Learning, and Natural Language Processing**

- Applied R to analyze financial time series data, optimize multi-asset investment portfolios using efficient frontier methods, and build logistic regression and classification tree models for credit risk and bankruptcy prediction.
- Built a panel regression model across 5,840 observations and 20 U.S. metro regions using interaction terms and Bayesian MCMC analysis to quantify a 97% decline in unemployment's sensitivity to housing prices from 2000 to 2024.
- Utilized TensorFlow and Keras to design neural network architectures for text classification, and fine-tuned FinBERT, a BERT-based transformer, to predict ESG scores from SEC 10-K filings by integrating NLP-derived sentiment features with structured financial data from the SEC EDGAR API.

### **Fantasy League Hub – Full-Stack Analytics Web Application**

- Built a Flask web application with 15+ routes, a Pandas data pipeline, and Plotly visualizations to deliver custom fantasy football analytics, engineering original metrics including Dominance Score, Luck Score, and Schedule Luck Rating from raw matchup and scoring data across 6 seasons.
- Designed and deployed a weighted Euclidean distance similarity engine with StandardScaler normalization and a logistic regression model predicting playoff qualification by quarter, translating outputs into accessible per-manager dashboards for a non-technical audience on a weekly in-season release schedule.

### **Sport Analytics Senior Thesis - Exploring the Financial Architecture of NBA Roster Construction Strategies**

- Led end-to-end data science project analyzing over 3,000 NBA player-seasons across 14 seasons, cleansing and integrating multi-source data from Basketball-Reference and RealGM to build three OLS regression models examining salary allocation, team playstyle, and offensive vs. defensive contributions to winning.
- Found that defensive production and ball control correlate more strongly with win percentage than offensive metrics, despite commanding lower salaries, with net defensive rating (+2.896% per unit) narrowly edging net offensive rating (+2.872%), while points and assists per game remain the primary drivers of individual contract size.

### **Syracuse University Basketball Analytics Competition**

- Collaborated in an agile team environment, competing against eight institutions to analyze NBA shot quality using clustering algorithms (K-Means) and sequential analysis on *Sports Info Solutions* data.
- Recognized as the winning team for exceptional teamwork and effective communication of technical findings to NBA team data and strategy analyst judges.

## Skills and Proficiencies

### **Technical & Analytical Tools**

- Python | SQL | R / R Studio | PySpark | Git | Tableau | Power BI | Excel | TensorFlow | Keras | scikit-learn | XGBoost | AWS | Stata | ggplot2

### **Data Science & Business Skills**

- Machine Learning | Statistical Modeling & Hypothesis Testing | OLS & Panel Regression | Data Visualization & Dashboard Development | NLP & Transformer Models | Exploratory Data Analysis | Logistic Regression & Classification Trees | Bayesian Analysis | ETL & Data Pipeline

